The Connes–Størmer Entropy of Two MASAs in the Two-by-Two Matrix Algebra: An Explicit Variational Formula

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Abstract

The relative entropy introduced by Connes and Størmer is defined by a supremum over all finite positive partitions of unity. For two maximal abelian subalgebras of the two-by-two complex matrix algebra, we prove that the unrestricted optimum lies in the orthogonal two-element family previously considered by Petz, Szántó and Weiner. We express the entropy as a one-dimensional moving-window maximum, identify the unique optimal angle for distinct subalgebras up to the natural symmetries, and obtain a cubic small-angle expansion. Its leading dependence on the relative angle is linear, whereas Choda's restricted entropy is quadratic up to a logarithmic factor. The proof combines antipodal symmetry of the Bloch ball with reduction to a great circle. In every matrix dimension greater than one, we also prove that Choda's partition is non-optimal for almost every unitary with respect to Haar measure. For one three-dimensional example, exact rational matrices and ball arithmetic certify a lower entropy bound from a four-element partition. Independent basis searches reproduce a candidate gap, but a budgeted global continuity-cover attempt does not bound all bases tightly enough. Its recorded obstruction leaves the comparison with the global basis optimum unresolved.

Keywords: Connes–Størmer entropy; relative position of subalgebras; maximal abelian subalgebras; unistochastic matrices; mutually unbiased bases; entropic uncertainty; partitions of unity.
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1 Introduction

1.1 An unrestricted partition problem

Let $\mathcal{M}$ be a finite von Neumann algebra with a faithful normal tracial state τ, and let $P, Q \subset \mathcal{M}$ be unital von Neumann subalgebras. Write E_P, E_Q for their trace-preserving conditional expectations and $\eta(t) = -t \log t$, with $\eta(0) = 0$. The relative entropy introduced by Connes and Størmer is

$$H(P|Q) = \sup_{\substack{x_j \geq 0 \\ \sum_j x_j = 1}} \sum_j \left[\tau\eta(E_Q x_j) - \tau\eta(E_P x_j)\right], \tag{1}$$

where the supremum is over finite partitions of arbitrary length [5, 18]. The definition permits noncommuting partition elements of arbitrary ranks. This freedom matters already when P and Q are two maximal abelian subalgebras (masas) of a matrix algebra.

Every pair of masas of $M_n = M_n(\mathbb{C})$ is unitarily equivalent to Δ_n and $Q_U = U\Delta_n U^*$, where Δ_n is diagonal and U is unitary. We use $\tau = \mathrm{Tr}\,/n$. The partition into the minimal projections of Δ_n gives

$$H(\Delta_n|Q_U) \geq h(U), \qquad h(U) = \frac{1}{n} \sum_{j,k} \eta(|u_{jk}|^2). \tag{2}$$

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Choda identified $h(U)$ with a modified entropy in which the admissible partitions are restricted to the first masa [3]. For complementary masas, $|u_{jk}|^2 = 1/n$, one has $h(U) = H = \log n$; conversely, $H = \log n$ characterises complementarity [17]. Outside this extremal case, restriction of the partitions generally changes the answer.

Petz, Szántó and Weiner proved strict inequality in (2) for a general pair of masas of M_2. They also introduced the one-parameter family of two-element orthogonal partitions that improves Choda's partition [17, Appendix]. Their construction supplies lower bounds. The question addressed here is whether any partition outside that family can do better. Our main result proves that none can, and locates the optimal member of the family. An arbitrary-length noncommutative optimisation therefore reduces to one real parameter without restricting the original admissible set by assumption.

1.2 The qubit formula and the optimal partition

For a state ρ put $G(\rho) = S(E_Q\rho) - S(E_P\rho)$, where $S(\rho) = \operatorname{Tr}\eta(\rho)$. The ensemble reformulation of Petz, Szántó and Weiner identifies (1) with the maximum average of G over ensembles of barycentre $1/n$. Thus H is the concave envelope of G at $1/n$, and $H \le \max G$. When $n = 2$, the reflection $\rho \mapsto 1 - \rho$ preserves the state space and leaves G invariant. A maximiser and its antipode average to $1/2$, proving $H = \max G$. This argument applies to any pair of subalgebras of M_2 (Lemma 3.2). For masas, a second argument shows that a maximiser is a pure state in the plane spanned by the two measurement axes.

Let $h_2(p) = -p\log p - (1-p)\log(1-p)$ and define the continuous periodic function

$$\psi(\beta) = \sin(2\beta)\log\tan^2\beta, \qquad \psi(\tfrac{\pi}{2}k) = 0 \quad (k \in \mathbb{Z}). \tag{3}$$

After removal of diagonal phases, a unitary in M_2 is the real rotation R_θ, where $|u_{11}|^2 = \cos^2\theta$. The following result describes the original unrestricted entropy.

Theorem A. *Let $U \in M_2$ be unitary, $c = |u_{11}|^2 = \cos^2\theta$ with $\theta \in [0, \frac{\pi}{2}]$, and*

$$\varphi(\alpha) = h_2\big(\cos^2(\alpha-\theta)\big) - h_2(\cos^2\alpha) = \int_{\alpha-\theta}^{\alpha} \psi(\beta)\,d\beta.$$

Then $H(\Delta_2|U\Delta_2U^*) = \max_{\alpha\in\mathbb{R}}\varphi(\alpha) = G^*$, where $G^* = \max_\rho G(\rho)$. The maximum is attained by the partition into the two projections onto an explicit orthonormal basis of $\mathbb{C}^2$, determined by a maximiser α^* of φ and the phases of U. For $\frac{1}{2} < c < 1$ the maximiser is unique modulo $\frac{\pi}{2}$, lies in $(\theta - \frac{\pi}{4}, 0)$ and is the unique solution there of $\psi(\alpha) = \psi(\alpha - \theta)$. The value of H is invariant under $c \mapsto 1 - c$ (Theorem 4.4 and Proposition 4.6).

The integral is a sliding-window formula: an optimal basis places a window of length θ over the positive hump of ψ, with equal values at the two endpoints. It is an explicit one-dimensional variational formula; no elementary expression for the optimal angle is asserted. Figure 2 illustrates this reduction, and the proof of Theorem 4.4 accounts for mixed states, additional partition elements and complex phases. The antipodal argument and the optimal-angle analysis promote the earlier family of lower bounds to an exact formula. The distinction from Choda's value is particularly pronounced for nearby masas.

Theorem B. *With the notation of Theorem A, $h(U) \le H(\Delta_2|U\Delta_2U^*) \le \kappa\min(\theta, \frac{\pi}{2} - \theta)$, where $\kappa = 2\sqrt{1-y^2}/y \approx 1.3255$ and $y \in (0,1)$ is the unique solution of $y\operatorname{artanh} y = 1$. As $\theta \to 0$,*

$$H(\Delta_2|U\Delta_2U^*) = \kappa\theta - \frac{\kappa + 4/\kappa}{6}\theta^3 + O(\theta^4), \qquad h(U) = 2\theta^2\log\frac{1}{\theta} + \theta^2 + O\Big(\theta^4\log\frac{1}{\theta}\Big)$$

(Theorem 5.1).

The leading term of H is linear in the relative angle, whereas $h(U)$ is quadratic with a logarithmic factor. Consequently $H/h(U)$ diverges as the masas coalesce. The first-order term is lost if one only tests partitions contained in the first masa. The result also improves entropy lower bounds for the spin-model subfactors considered by Bakshi and Guin [1]; the direct application is stated in Remark 5.3.

1.3 What survives in higher dimension

For $n \geq 3$, reflection through the maximally mixed state no longer preserves the state space: the image of a pure state under $\rho \mapsto 2/n - \rho$ has the negative eigenvalue $2/n - 1$. Hence the qubit proof gives no basis for identifying H with $\max G$, or for restricting optimal partitions to orthonormal bases. We prove instead a first-order criterion. If

$$M_U = (u_{jk} \log |u_{jk}|^2)_{jk}\, U^*$$

is not Hermitian, an explicit infinitesimal rotation of Choda's basis increases the partition value (Proposition 6.1). This condition holds on a nonempty open set of $U(n)$ in every dimension. A connectedness and real-analytic zero-set argument then proves strict improvement for Haar-almost every unitary (Theorem 6.5), including dimensions greater than two.

The historical study in Section 6 compares local searches over general partitions, bases and single states for $n = 3, 4, 5$. Some explicit partitions have larger recomputed values than every basis found by extended multistart searches. This motivates a strict separation between the two optimisation problems, but does not prove such a separation: no global upper bound for the basis problem was obtained. High-precision evaluation of a candidate partition is also distinct from a rigorous numerical error enclosure. For one explicit three-dimensional unitary we now provide exact rational point witnesses and rigorously enclose their entropies. An independent basis/frame search and a budgeted global continuity-cover attempt are reported in Section 8, including the numerical reason the attempted global cover does not close the gap.

1.4 Relation to earlier work

Connes–Størmer entropy and its connection with index theory are developed in [5, 18, 9, 14, 13]. Popa's orthogonality of subalgebras [19] provides the maximal-entropy complementary case; for masas this is the setting of mutually unbiased bases and complex Hadamard matrices [16, 24, 6, 21]. The matrix of squared moduli in (2) is unistochastic, and its geometry is already nontrivial for $n = 3$ [2].

The ensemble representation, the previously used one-parameter family, strict non-optimality of Choda's partition in M_2, and the characterisation of complementary masas are credited to their original sources [3, 17]. Our contribution is global optimality of that family, the location of its optimal parameter, the qubit asymptotics, the higher-dimensional first-order criterion and its almost-everywhere consequence. The concave-envelope bound and its information-theoretic interpretation are elementary consequences included to explain the proof. For masas the entropy maximises a difference of the information supplied by two measurements of an ensemble of average $1/n$ (Proposition 2.4), complementing the sum bounds in entropic uncertainty and information exclusion [11, 7, 4].

1.5 Structure and numerical provenance

Sections 2 and 3 establish the ensemble and symmetry arguments. Sections 4 and 5 prove the qubit formula, uniqueness and asymptotic expansion. Section 6 separates the higher-dimensional theorem from exploratory numerical observations. Section 7 discusses their implications and Section 8 reports the new attempted gap certification, including a completed pointwise lower bound and the failed global covering step; Appendix A records the numerical method.

All analytic results have proofs in the text and do not depend on the numerical searches. The stored partition-search data were produced with revision `e8d5873` of the accompanying code [10]. The tables and figures reproduce those historical outputs; preparing this manuscript has not repeated the full-size study after the subsequent numerical corrections to the library. The data and figure assets are included locally, and the source repository records the reproduction status in `man/REPRODUCIBILITY.md`.

The new computations in Section 8 are separate from those historical tables. They use an independent NumPy/SciPy parameterisation and an exact rational/Arb point checker; the supplied scripts and JSON record the fixed seeds, optimisation budgets, exact witness parameters and covering attempt.

2 Preliminaries

2.1 Conditional expectations and the entropy

Throughout, $\mathcal{M} = M_n$, $\tau = \operatorname{Tr}/n$, and $\langle x, y\rangle_\tau = \tau(x^*y)$. Subalgebras are unital $*$-subalgebras. For such a subalgebra P the τ-preserving conditional expectation E_P is the orthogonal projection of $(\mathcal{M}, \langle\cdot,\cdot\rangle_\tau)$ onto P; it is unital, positive, P-bimodular, preserves τ and Tr, and is the unique conditional expectation onto P with these properties (see e.g. [22]). The function $\eta(t) = -t\log t$, $\eta(0) = 0$, is continuous and concave on $[0,\infty)$; for $x \geq 0$, $\eta(x)$ is defined by functional calculus, and $S(\sigma) = \operatorname{Tr}\eta(\sigma)$ is the von Neumann entropy of a density matrix σ. The state space $\mathcal{D} = \{\rho \in M_n : \rho \geq 0,\ \operatorname{Tr}\rho = 1\}$ is a compact convex set of real dimension $n^2 - 1$. We write $h_2(p) = \eta(p) + \eta(1-p)$ for the binary entropy, $p \in [0,1]$, and reserve the letter h for Choda's value (2), which we regard as a function on all unitaries.

Definition 2.1. For subalgebras $P, Q \subset \mathcal{M}$ and $x \geq 0$ put $F(x) = F_{P,Q}(x) = \tau\eta(E_Q x) - \tau\eta(E_P x)$. The Connes–Størmer relative entropy is $H(P|Q) = \sup\sum_j F(x_j)$ over all finite families $x_1, \ldots, x_m \geq 0$ in $\mathcal{M}$ with $\sum_j x_j = 1$ (partitions of unity).

Lemma 2.2. *Let $P, Q \subset \mathcal{M}$ be subalgebras.*

(a) $F(\lambda x) = \lambda F(x)$ for $x \geq 0$ and $\lambda \geq 0$. In particular, merging two elements of a partition that are multiples of the same positive operator does not change $\sum_j F(x_j)$.

(b) Let $v \in \mathcal{M}$ be unitary and $\gamma = \operatorname{Ad} v$. Then $E_{\gamma(B)} = \gamma \circ E_B \circ \gamma^{-1}$ for every subalgebra B, $F_{\gamma(P),\gamma(Q)}(\gamma(x)) = F_{P,Q}(x)$, and $H(\gamma(P)|\gamma(Q)) = H(P|Q)$; γ maps optimal partitions for (P, Q) onto optimal partitions for $(\gamma(P), \gamma(Q))$.

Proof. (a) For $s, \lambda \geq 0$, $\eta(\lambda s) = \lambda\eta(s) + s\,\eta(\lambda)$, hence $\tau\eta(\lambda y) = \lambda\tau\eta(y) + \eta(\lambda)\tau(y)$ for $y \geq 0$. Since $\tau(E_Q x) = \tau(E_P x) = \tau(x)$, the terms with $\eta(\lambda)$ cancel in $F(\lambda x)$. If $x_1 = a\,y$ and $x_2 = b\,y$ with $a, b \geq 0$, then $F(x_1) + F(x_2) = (a+b)F(y) = F(x_1 + x_2)$.

(b) γ is unitary for $\langle\cdot,\cdot\rangle_\tau$, since $\tau(\gamma(x)^*\gamma(y)) = \tau(vx^*yv^*) = \tau(x^*y)$, and $\gamma(B)$ is the image of B; so $\gamma E_B \gamma^{-1}$ is the orthogonal projection onto $\gamma(B)$. Moreover $\eta(\gamma(y)) = \gamma(\eta(y))$ and $\tau\circ\gamma = \tau$, so $\tau\eta(E_{\gamma(B)}\gamma(x)) = \tau\gamma(\eta(E_B x)) = \tau\eta(E_B x)$. Since γ maps partitions of unity bijectively onto partitions of unity, the remaining claims follow. □

2.2 Two masas

Let Δ_n be the diagonal masa and $Q_U = U\Delta_n U^*$ for a unitary U with columns $u_1, \ldots, u_n$. The minimal projections of Q_U are $p_k = u_k u_k^*$, and

$$E_{\Delta_n}(x) = \sum_k e_{kk} x e_{kk}, \qquad E_{Q_U}(x) = \sum_k p_k x p_k = \sum_k \langle u_k, x u_k\rangle\, p_k, \tag{4}$$

since $\sum_k p_k x p_k \in Q_U$ and $\tau(p_l(x - \sum_k p_k x p_k)) = \tau(p_l x) - \tau(p_l x p_l) = 0$ for every l. Thus the spectrum of $E_{Q_U} x$ is the diagonal of $U^* x U$, and

$$F(x) = \frac{1}{n}\Big[\sum_k \eta((U^* x U)_{kk}) - \sum_k \eta(x_{kk})\Big]. \tag{5}$$

The algebra Q_U does not change if U is multiplied from the right by a diagonal unitary or a permutation matrix, and by Lemma 2.2(b) with $\gamma = \operatorname{Ad} D$ for a diagonal unitary D (which fixes Δ_n), $H(\Delta_n|Q_{DU}) = H(\Delta_n|Q_U)$. For the partition $(e_{kk})_k$, $\eta(x_{kk}) \in \{\eta(0), \eta(1)\} = \{0\}$ and $(U^* e_{kk} U)_{jj} = |u_{kj}|^2$, which gives (2). For an orthonormal basis $w_1, \dots, w_n$, the columns of a unitary W, the partition $x_j = w_j w_j^*$ gives, by (5),

$$\sum_j F(w_j w_j^*) = h(U^* W) - h(W), \tag{6}$$

and Choda's partition is the case $W = 1$. We call (6) the value of the basis W.

2.3 The ensemble form

The following reformulation is due to Petz, Szántó and Weiner [17, Section 3, Lemma 1 and (7), (8)]; we include the short proof, and a Carathéodory argument for the attainment of the supremum.

Proposition 2.3 (Petz–Szántó–Weiner)**.** *Let $P, Q \subset M_n$ be subalgebras and $G(\rho) = S(E_Q\rho) - S(E_P\rho)$ for $\rho \in \mathcal{D}$.*

(a) A partition (x_j) with $x_j \neq 0$ corresponds to the ensemble $t_j = \tau(x_j)$, $\rho_j = x_j/\operatorname{Tr} x_j$, which satisfies $t_j > 0$, $\sum_j t_j = 1$ and $\sum_j t_j \rho_j = \frac{1}{n}$; conversely every such ensemble defines the partition $x_j = n t_j \rho_j$. In both directions $\sum_j F(x_j) = \sum_j t_j G(\rho_j)$.

(b) $H(P|Q) = \max\{\sum_j t_j G(\rho_j) :\ t_j \geq 0,\ \sum_j t_j = 1,\ \sum_j t_j \rho_j = \frac{1}{n}\}$; the maximum is attained by an ensemble of at most $n^2 + 1$ states. (Petz, Szántó and Weiner show that n^2 linearly independent states suffice.)

(c) $G \leq \log n$ on $\mathcal{D}$, hence $H(P|Q) \leq \log n$.

Proof. (a) Elements $x_j = 0$ contribute $F(0) = 0$ and can be dropped. $\operatorname{Tr} x_j = n\tau(x_j) = n t_j$, so $x_j = n t_j \rho_j$, and $\sum_j x_j = 1$ is equivalent to $\sum_j t_j \rho_j = \frac{1}{n}$, which implies $\sum_j t_j = 1$ by taking traces. Since $E_P\rho$ and $E_Q\rho$ are density matrices, $F(\rho) = \frac{1}{n}[S(E_Q\rho) - S(E_P\rho)] = \frac{1}{n}G(\rho)$, and Lemma 2.2(a) gives $F(x_j) = n t_j F(\rho_j) = t_j G(\rho_j)$.

(b) By (a), H is the supremum of $\sum_j t_j G(\rho_j)$ over finite ensembles with barycentre $\frac{1}{n}$. Identify the affine hull of $\mathcal{D}$ with $\mathbb{R}^{n^2-1}$. For such an ensemble the point $(\frac{1}{n}, \sum_j t_j G(\rho_j)) \in \mathbb{R}^{n^2}$ is a convex combination of points of the graph $\{(\rho, G(\rho)) : \rho \in \mathcal{D}\}$; by Carathéodory's theorem it is a convex combination of at most $n^2 + 1$ of them, i.e. there is an ensemble of at most $n^2 + 1$ states with the same barycentre and the same value. The ensembles of $n^2 + 1$ states (zero weights allowed) with barycentre $\frac{1}{n}$ form a compact set, on which $\sum_j t_j G(\rho_j)$ is continuous, because S is continuous on $\mathcal{D}$ and E_P, E_Q are linear. So the supremum is a maximum.

(c) $S(E_Q\rho) \leq \log n$ for the density matrix $E_Q\rho$ of M_n, and $S(E_P\rho) \geq 0$. □

For two masas, $G(\rho) = H_{\mathrm{Sh}}(\operatorname{diag}(U^*\rho U)) - H_{\mathrm{Sh}}(\operatorname{diag}\rho)$ with the Shannon entropy H_{Sh} of a probability vector, by (4).

2.4 The information-theoretic reading

For an ensemble $\{t_j, \rho_j\}$ and a subalgebra $P \subset M_n$ let

$$\chi_P\{t_j,\rho_j\} = S\Big(\sum_j t_j E_P\rho_j\Big) - \sum_j t_j S(E_P\rho_j)$$

be the Holevo quantity of the ensemble $\{t_j, E_P\rho_j\}$, i.e. of the ensemble after the quantum channel E_P. If P is the masa of the minimal projections $p_1,\dots,p_n$, then $E_P\rho_j = \sum_k \operatorname{Tr}(p_k\rho_j)\,p_k$ by (4), and χ_P is the classical mutual information $I(J : X_P)$ between a label J with distribution (t_j) and the outcome X_P of the measurement (p_k) on ρ_J, whose conditional distribution is $\Pr(X_P = k \,|\, J = j) = \operatorname{Tr}(p_k\rho_j)$.

Proposition 2.4. *Let $P, Q \subset M_n$ be subalgebras and $\{t_j, \rho_j\}$ an ensemble with $\sum_j t_j\rho_j = \frac{1}{n}$. Then*

$$\sum_j t_j G(\rho_j) = \chi_P\{t_j,\rho_j\} - \chi_Q\{t_j,\rho_j\}, \qquad \textit{hence} \qquad H(P|Q) = \max(\chi_P - \chi_Q)$$

over these ensembles. For $P = \Delta_n$ and $Q = U\Delta_n U^*$ this reads $\sum_j t_j G(\rho_j) = I(J : X_P) - I(J : X_Q)$, where X_P and X_Q are the outcomes of measuring ρ_J in the standard basis and in the basis $u_1,\dots,u_n$.

Proof. E_P is linear and unital, so $\sum_j t_j E_P\rho_j = E_P(\frac{1}{n}) = \frac{1}{n}$ and $S(\sum_j t_j E_P\rho_j) = \log n$; the same holds for Q. Hence $\chi_P - \chi_Q = \sum_j t_j\,[S(E_Q\rho_j) - S(E_P\rho_j)] = \sum_j t_j G(\rho_j)$, and Proposition 2.3(b) gives the formula for H. The last statement is the identification $\chi_P = I(J : X_P)$ above. □

In this language Choda's partition is the ensemble of the standard basis vectors with equal weights, for which $X_P = J$, $I(J : X_P) = \log n$ and $I(J : X_Q) = \log n - h(U)$, and the complementary case is the one in which this ensemble is completely invisible to the second measurement. Since the Shannon entropy of a probability vector is at least the negative logarithm of its largest entry, $h(U) \geq -\log \max_{i,j} |u_{ij}|^2$; this is the bound of Maassen and Uffink [11] for $S(E_P\rho) + S(E_Q\rho)$, evaluated at the basis vectors $\rho = e_{kk}$, where $S(E_P\rho) = 0$.

3 The concave envelope and the antipodal symmetry

For a bounded function g on $\mathcal{D}$ let $\hat{g}(\rho) = \sup\{\sum_j t_j g(\rho_j) :$ finite ensembles with $\sum_j t_j\rho_j = \rho\}$; this is the smallest concave function on $\mathcal{D}$ that majorises g (its concave envelope).

Proposition 3.1. *Let $P, Q \subset M_n$ be subalgebras, G as in Proposition 2.3, $G^* = \max_{\mathcal{D}} G$ and $A = \{\rho \in \mathcal{D} : G(\rho) = G^*\}$. Then*

$$H(P|Q) = \hat{G}(\tfrac{1}{n}) \leq G^*,$$

and $H(P|Q) = G^*$ if and only if $\frac{1}{n} \in \operatorname{conv} A$. In that case the optimal partitions are exactly the partitions $x_j = nt_j\rho_j$ with $\rho_j \in A$ whenever $t_j > 0$.

Proof. The identity $H = \hat{G}(\frac{1}{n})$ is Proposition 2.3(b) with the definition of $\hat{G}$. For every ensemble, $\sum_j t_j G(\rho_j) \leq \sum_j t_j G^* = G^*$, with equality if and only if $G(\rho_j) = G^*$ whenever $t_j > 0$. If $\frac{1}{n} = \sum_j t_j\rho_j$ with $\rho_j \in A$, this ensemble has the value G^*, so $H = G^*$. Conversely, if $H = G^*$, the maximum in Proposition 2.3(b) is attained by an ensemble with value G^*, all of whose states with positive weight lie in A; hence $\frac{1}{n} \in \operatorname{conv} A$. The last statement is the equality case together with Proposition 2.3(a). □

The bound G^* is a maximisation over the (n^2-1)-dimensional state space, whereas H requires ensembles of up to n^2 states with a linear constraint; any state gives a lower bound for G^*, any partition a lower bound for H. Upper bounds for H come from the dual side: if $a(\rho) = a_0 + \mathrm{Tr}(A\rho)$ is an affine function with $a \geq G$ on $\mathcal{D}$, then $\sum_j t_j G(\rho_j) \leq a(\sum_j t_j \rho_j)$ for every ensemble, and so

$$H(P|Q) \leq a(\tfrac{1}{n}) = a_0 + \tfrac{1}{n} \operatorname{Tr} A. \tag{7}$$

Since $\hat{G}$ is concave and upper semicontinuous (the maximum in Proposition 2.3(b) is over a compact set), it is the infimum of its affine majorants [20, Section 12], so these bounds are sharp in the limit. The constant function $a = G^*$ gives Proposition 3.1.

The equality case of Proposition 3.1 is exactly what a symmetry of the state space can provide.

Lemma 3.2 (Antipodal symmetry)**.** *Let $n = 2$ and let $P, Q \subset M_2$ be subalgebras. The map $\rho \mapsto 1 - \rho$ is an affine involution of $\mathcal{D}$ that fixes $\frac{1}{2}$ and satisfies $G(1-\rho) = G(\rho)$. Consequently $H(P|Q) = G^*$, and for every maximiser ρ^* of G the partition $\{\rho^*, 1-\rho^*\}$ is optimal. In the Bloch parametrisation $\rho = \frac{1}{2}(1 + r \cdot \sigma)$ the map is $r \mapsto -r$.*

Proof. If $\rho \geq 0$ and $\operatorname{Tr}\rho = 1$, the eigenvalues $\lambda, 1-\lambda \in [0,1]$ of ρ give $1-\rho \geq 0$, and $\mathrm{Tr}(1-\rho) = 1$. Since E_P is linear and unital, $E_P(1-\rho) = 1 - E_P\rho$, and for a 2×2 density matrix σ with eigenvalues $\mu, 1-\mu$ the matrix $1-\sigma$ has the eigenvalues $1-\mu, \mu$, so $S(1-\sigma) = S(\sigma)$. Hence $S(E_P(1-\rho)) = S(E_P\rho)$, likewise for Q, and $G(1-\rho) = G(\rho)$. If $\rho^* \in A$, then $1-\rho^* \in A$ and $\frac{1}{2} = \frac{1}{2}\rho^* + \frac{1}{2}(1-\rho^*) \in \operatorname{conv} A$, so $H = G^*$ by Proposition 3.1, with the optimal partition $x_\pm = 2 \cdot \frac{1}{2}\rho_\pm$, $\rho_+ = \rho^*$, $\rho_- = 1-\rho^*$. □

Remark 3.3. (i) For $n \geq 3$ the point reflection $\rho \mapsto \frac{2}{n} - \rho$ through the centre of $\mathcal{D}$ is affine and fixes $\frac{1}{n}$, but it does not preserve $\mathcal{D}$: a pure state is mapped to an operator with the eigenvalue $\frac{2}{n} - 1 < 0$. Nothing forces $\frac{1}{n} \in \operatorname{conv} A$, and for $n = 3, 4, 5$ the state searches of Section 6 return values above the best partition values found. This suggests a strict gap $H < G^*$ but proves neither that gap nor any statement about exact optimal ensembles. (ii) If U is complementary, $G(e_{11}) = \log n = G^*$ by (5), and $\frac{1}{n} = \frac{1}{n}\sum_k e_{kk}$ is a convex combination of the maximisers e_{kk} (all of which satisfy $G(e_{kk}) = \log n$); so $H = G^* = \log n$, the partition (e_{kk}) is optimal, and $h(U) = \log n$.

4 Two masas of M_2

4.1 The function ψ

Let $\Psi(\beta) = h_2(\cos^2\beta)$ and let ψ be as in (3). Throughout, β denotes the variable of ψ and Ψ.

Lemma 4.1. *(a) ψ is continuous on $\mathbb{R}$, odd, of period $\frac{\pi}{2}$, vanishes exactly on $\frac{\pi}{4}\mathbb{Z}$, and $\psi > 0$ on $(-\frac{\pi}{4}, 0)$, $\psi < 0$ on $(0, \frac{\pi}{4})$.*

(b) Ψ is continuously differentiable on $\mathbb{R}$, $\Psi' = -\psi$, and Ψ is even and of period $\frac{\pi}{2}$.

(c) On $(-\frac{\pi}{4}, 0)$, ψ has a unique critical point β_κ; ψ is strictly increasing on $[-\frac{\pi}{4}, \beta_\kappa]$ and strictly decreasing on $[\beta_\kappa, 0]$. With $\beta_\kappa = -s^*/2$ and $y = \cos s^*$, $y \in (0,1)$ is the unique solution of $y \operatorname{artanh} y = 1$, and

$$\kappa := \max_{\mathbb{R}} \psi = \psi(\beta_\kappa) = 2\tan s^* = \frac{2\sqrt{1-y^2}}{y}, \qquad \psi''(\beta_\kappa) = -4\kappa - \frac{16}{\kappa}.$$

Proof. (a) For $\beta \notin \frac{\pi}{2}\mathbb{Z}$, $\psi(-\beta) = -\psi(\beta)$ since sin is odd and $\tan^2$ even, and $\psi(\beta + \frac{\pi}{2}) = \psi(\beta)$ since $\sin(2\beta + \pi) = -\sin 2\beta$ and $\tan^2(\beta + \frac{\pi}{2}) = \cot^2\beta$, so that $\log\tan^2(\beta + \frac{\pi}{2}) = -\log\tan^2\beta$. Near 0, $|\psi(\beta)| \leq 2|\beta|\,|\log\tan^2\beta| \to 0$, so ψ is continuous at 0 with $\psi(0) = 0$, and by periodicity at every point of $\frac{\pi}{2}\mathbb{Z}$. On $(0, \frac{\pi}{4})$, $\sin 2\beta > 0$ and $0 < \tan^2\beta < 1$, so $\psi < 0$; by oddness $\psi > 0$ on $(-\frac{\pi}{4}, 0)$. $\psi(\pm\frac{\pi}{4}) = \pm\log 1 = 0$. By periodicity these are all zeros.

(b) h_2 is smooth on $(0,1)$ with $h_2'(p) = \log\frac{1-p}{p}$. For $\beta \notin \frac{\pi}{2}\mathbb{Z}$, $0 < \cos^2\beta < 1$ and $\Psi'(\beta) = h_2'(\cos^2\beta)\cdot(-\sin 2\beta) = -\sin 2\beta \log\tan^2\beta = -\psi(\beta)$. At $\beta_0 \in \frac{\pi}{2}\mathbb{Z}$, Ψ is continuous, and by the mean value theorem $(\Psi(\beta)-\Psi(\beta_0))/(\beta-\beta_0) = -\psi(\xi)$ for some ξ between β_0 and β; by continuity of ψ this tends to $-\psi(\beta_0) = 0$. Hence $\Psi \in C^1(\mathbb{R})$, $\Psi' = -\psi$. Evenness and periodicity follow from $\cos^2(-\beta) = \cos^2\beta$ and $h_2(\cos^2(\beta+\frac{\pi}{2})) = h_2(\sin^2\beta) = h_2(\cos^2\beta)$.

(c) Away from $\frac{\pi}{2}\mathbb{Z}$, $(\log\tan^2\beta)' = 2/(\sin\beta\cos\beta) = 4/\sin 2\beta$, hence

$$\psi'(\beta) = 2\cos 2\beta \log\tan^2\beta + 4, \qquad \psi''(\beta) = -4\psi(\beta) + 8\cot 2\beta.$$

For $b \in (0,\frac{\pi}{4})$ let $g(b) = \cos 2b \log\tan b$, so that $\psi'(-b) = 4(g(b)+1)$. Then $g'(b) = -2\sin 2b\log\tan b + 2\cos 2b/\sin 2b > 0$, $g(b) \to -\infty$ as $b \to 0$ and $g(\frac{\pi}{4}) = 0$. Hence $g = -1$ at exactly one point b^*, $\psi' > 0$ on $(-\frac{\pi}{4}, -b^*)$ and $\psi' < 0$ on $(-b^*, 0)$; put $\beta_\kappa = -b^*$, $s^* = 2b^* \in (0,\frac{\pi}{2})$. Since $\log\tan b = -\operatorname{artanh}(\cos 2b)$ (both sides equal $\frac{1}{2}\log\frac{1-\cos 2b}{1+\cos 2b}$), the equation $g(b^*) = -1$ reads $y\operatorname{artanh} y = 1$ with $y = \cos s^*$, and $y\operatorname{artanh} y$ increases strictly from 0 to ∞ on $(0,1)$. Then $\psi(\beta_\kappa) = -\sin s^* \cdot 2\log\tan b^* = 2\sin s^*/\cos s^* = 2\tan s^*$. By (a) the maximum of ψ over a period is attained in $[-\frac{\pi}{4}, 0]$, hence at β_κ. Finally $\psi''(\beta_\kappa) = -4\kappa + 8\cot(-s^*) = -4\kappa - 8/\tan s^* = -4\kappa - 16/\kappa$. □

Numerically $\kappa = 1.3255$ and $\beta_\kappa \approx -0.2926$.

4.2 The variational formula

Write $R_\theta = \left(\begin{smallmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{smallmatrix}\right)$.

Lemma 4.2. *Every unitary $U \in M_2$ can be written as $U = D_1 R_\theta D_2$ with diagonal unitaries D_1, D_2 and $\theta \in [0,\frac{\pi}{2}]$, $\cos^2\theta = |u_{11}|^2$. Consequently $H(\Delta_2|U\Delta_2U^*) = H(\Delta_2|R_\theta\Delta_2R_\theta^T)$, and $\gamma = \operatorname{Ad} D_1$ maps optimal partitions for R_θ onto optimal partitions for U.*

Proof. The columns and the rows of U are unit vectors, so $|u_{11}|^2 + |u_{21}|^2 = 1 = |u_{11}|^2 + |u_{12}|^2$ and $|u_{12}|^2 + |u_{22}|^2 = 1$; hence $|u_{21}| = |u_{12}|$ and $|u_{22}| = |u_{11}|$. Let $\theta \in [0,\frac{\pi}{2}]$ with $\cos\theta = |u_{11}|$ and write $u_{11} = e^{ia}\cos\theta$, $u_{21} = e^{ib}\sin\theta$, $u_{12} = -e^{ic}\sin\theta$, $u_{22} = e^{id}\cos\theta$. If $0 < \theta < \frac{\pi}{2}$, orthogonality of the columns gives $(e^{i(d-b)} - e^{i(c-a)})\cos\theta\sin\theta = 0$, i.e. $e^{i(d-b)} = e^{i(c-a)}$, and then $U = \operatorname{diag}(e^{ia}, e^{ib})\, R_\theta\, \operatorname{diag}(1, e^{i(c-a)})$, as one checks entry by entry. If $\theta = 0$, U is diagonal and $U = U\cdot R_0\cdot 1$; if $\theta = \frac{\pi}{2}$, $U = \operatorname{diag}(-u_{12}, u_{21})R_{\pi/2}$. Since $D_2\Delta_2D_2^* = \Delta_2$, $U\Delta_2U^* = \gamma(R_\theta\Delta_2R_\theta^T)$ with $\gamma = \operatorname{Ad} D_1$, which fixes Δ_2; Lemma 2.2(b) gives the rest. □

We use the Pauli matrices $\sigma = (\sigma_1,\sigma_2,\sigma_3)$ and the Bloch parametrisation $\rho = \frac{1}{2}(1 + r\cdot\sigma)$, $r \in \mathbb{R}^3$, $|r| \le 1$, of $\mathcal{D}$ for $n = 2$; ρ is a rank-one projection if and only if $|r| = 1$. For a unit vector $v = (\cos a, \sin a)^T$ one has $vv^* = \frac{1}{2}(1 + \sin 2a\,\sigma_1 + \cos 2a\,\sigma_3)$, so its Bloch vector is $(\sin 2a, 0, \cos 2a)$, and the Bloch vector of $(-\sin a, \cos a)^T$ is the negative of it. Let

$$z = (0,0,1), \qquad q = (\sin 2\theta, 0, \cos 2\theta), \qquad k(s) = h_2\big(\tfrac{1+s}{2}\big)\ (|s| \le 1),$$

so z and q are the Bloch vectors of e_1 and of $R_\theta e_1$, and k is even.

Lemma 4.3. *For $P = \Delta_2$, $Q = R_\theta\Delta_2R_\theta^T$ and $\rho = \frac{1}{2}(1 + r\cdot\sigma)$, $G(\rho) = f(r) := k(r\cdot q) - k(r\cdot z)$. Moreover*

$$k(s) = \log 2 - \sum_{m\ge1} \frac{s^{2m}}{2m(2m-1)} \qquad (|s| \le 1), \tag{8}$$

the series converging uniformly on $[-1,1]$.

Proof. $E_{\Delta_2}\rho = \operatorname{diag}(\rho)$ has the eigenvalues $\frac{1}{2}(1 \pm r_3) = \frac{1}{2}(1 \pm r\cdot z)$, so $S(E_{\Delta_2}\rho) = k(r\cdot z)$. By (4) the eigenvalues of $E_Q\rho$ are $\langle v_1, \rho v_1\rangle$ and $\langle v_2, \rho v_2\rangle = 1 - \langle v_1, \rho v_1\rangle$ with $v_1 = R_\theta e_1$, and $\langle v_1, \rho v_1\rangle = \operatorname{Tr}(\rho v_1 v_1^*) = \frac{1}{2}(1 + r\cdot q)$ by $\operatorname{Tr}((1+r\cdot\sigma)(1+q\cdot\sigma)) = 2(1+r\cdot q)$. So $S(E_Q\rho) = k(r\cdot q)$.

For (8), $k(s) = \log 2 - \frac{1}{2}\big[(1+s)\log(1+s) + (1-s)\log(1-s)\big]$. For $|s| < 1$, integrating $\frac{d^2}{ds^2}(1 \pm s)\log(1 \pm s) = 1/(1 \pm s) = \sum_{j\geq 0}(\mp s)^j$ twice gives $(1 \pm s)\log(1 \pm s) = \pm s + \sum_{l\geq 2}(\mp 1)^l s^l/(l(l-1))$, and adding the two expansions leaves $2\sum_{m\geq 1} s^{2m}/(2m(2m-1))$. The terms are bounded by $1/(2m(2m-1))$ on $[-1,1]$, so the series converges uniformly there (Weierstrass), its sum is continuous, and the identity extends to $s = \pm 1$ by continuity of k. □

Theorem 4.4. *Let $U \in M_2$ be unitary, $U = D_1 R_\theta D_2$ as in Lemma 4.2, $c = |u_{11}|^2 = \cos^2\theta$, and*

$$\varphi(\alpha) = h_2(\cos^2(\alpha-\theta)) - h_2(\cos^2\alpha) = \int_{\alpha-\theta}^{\alpha} \psi(\beta)\, d\beta.$$

Then φ is C^1 and $\frac{\pi}{2}$-periodic, $\varphi'(\alpha) = \psi(\alpha) - \psi(\alpha-\theta)$, and

(a) $H(\Delta_2|U\Delta_2U^*) = \max_{\alpha\in\mathbb{R}} \varphi(\alpha) = G^*$;

(b) for every maximiser α^* of φ, the projections onto the orthonormal basis $D_1(\cos\alpha^*, \sin\alpha^*)^T$, $D_1(-\sin\alpha^*, \cos\alpha^*)^T$ form an optimal partition;

(c) $H(\Delta_2|U\Delta_2U^*)$ depends only on c, and it is invariant under $c \mapsto 1-c$.

Proof. By Lemma 4.2 we may assume $U = R_\theta$, $D_1 = 1$, and transport optimal partitions by $\mathrm{Ad}\, D_1$ at the end. The integral form is $\varphi(\alpha) = \Psi(\alpha-\theta) - \Psi(\alpha) = \int_{\alpha-\theta}^{\alpha}\psi$ by Lemma 4.1(b), which also gives the regularity, periodicity and derivative of φ.

Step 1: $H = G^* = \max_{|r|\leq 1} f(r)$. This is Lemma 3.2 together with Lemma 4.3; in the Bloch picture, $f(-r) = f(r)$ because k is even, and if r^* maximises f, the partition $\{\rho_+, \rho_-\}$ with $\rho_\pm = \frac{1}{2}(1 \pm r^*\cdot\sigma)$ is optimal.

Step 2: reduction to the unit circle in the plane of z and q. If $\theta \in \{0, \frac{\pi}{2}\}$, then $q = \pm z$ and $f \equiv 0$ since k is even; also $\varphi \equiv 0$, because $h_2(\cos^2(\alpha - \frac{\pi}{2})) = h_2(\sin^2\alpha) = h_2(\cos^2\alpha)$. So $H = 0 = \max\varphi$ and every basis is optimal. Let $0 < \theta < \frac{\pi}{2}$. Then z and q span the plane $\{r_2 = 0\}$, and $f(r)$ depends only on the orthogonal projection $a = (r_1, 0, r_3)$ of r onto it, $|a| \leq |r| \leq 1$. Write $a = \varrho(\sin\omega, 0, \cos\omega)$ with $0 \leq \varrho \leq 1$ and the polar angle ω measured from z; then $a\cdot z = \varrho\cos\omega$ and $a\cdot q = \varrho\cos(\omega - 2\theta)$, and by (8)

$$f(a) = \sum_{m\geq 1} \frac{\varrho^{2m}}{2m(2m-1)}\big(\cos^{2m}\omega - \cos^{2m}(\omega - 2\theta)\big).$$

If $|\cos\omega| \geq |\cos(\omega-2\theta)|$, all terms are non-negative and non-decreasing in ϱ, so $f(a) \leq f(\sin\omega, 0, \cos\omega)$. Otherwise all terms are non-positive and $f(a) \leq 0 = f(\sin\theta, 0, \cos\theta)$, the value at $\omega = \theta$, where $\cos\omega = \cos(\omega - 2\theta)$. Hence the maximum of f over the ball equals its maximum over the unit circle of the plane,

$$\max_{|r|\leq 1} f = \max_{\omega\in\mathbb{R}}\big[k(\cos(\omega-2\theta)) - k(\cos\omega)\big],$$

and it is attained at a unit vector $r^* = (\sin\omega^\circ, 0, \cos\omega^\circ)$.

Step 3: the angle. Since $k(\cos 2\gamma) = h_2(\frac{1+\cos 2\gamma}{2}) = h_2(\cos^2\gamma)$, the substitution $\omega = 2\alpha$ turns the right-hand side into $\max_\alpha \varphi(\alpha)$. This proves (a). For a maximiser α^* of φ, $r^* = (\sin 2\alpha^*, 0, \cos 2\alpha^*)$ maximises f, and ρ_+, ρ_- are the projections onto $(\cos\alpha^*, \sin\alpha^*)^T$ and $(-\sin\alpha^*, \cos\alpha^*)^T$; with Step 1 and Lemma 4.2 this proves (b).

(c) φ depends only on θ, i.e. on c. Let $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Then $UJ\Delta_2(UJ)^* = U\Delta_2U^*$ and $|(UJ)_{11}|^2 = |u_{12}|^2 = 1 - c$. □

Remark 4.5. (i) At $\alpha = 0$, $\varphi(0) = h_2(c) = h(U)$, the value of Choda's partition e_{11}, e_{22}. At $\alpha = \theta$ (the basis of Q) $\varphi(\theta) = -h(U)$. (ii) In the Bloch picture (Figure 1), z and q enclose the angle 2θ, and the optimal Bloch vector r^* has the angle $2\alpha^*$ from z. By Proposition 4.6 below, for $\frac{1}{2} < c < 1$ it lies on the other side of z than q. (iii) Let $0 < c < 1$. Since $H = G^*$, the

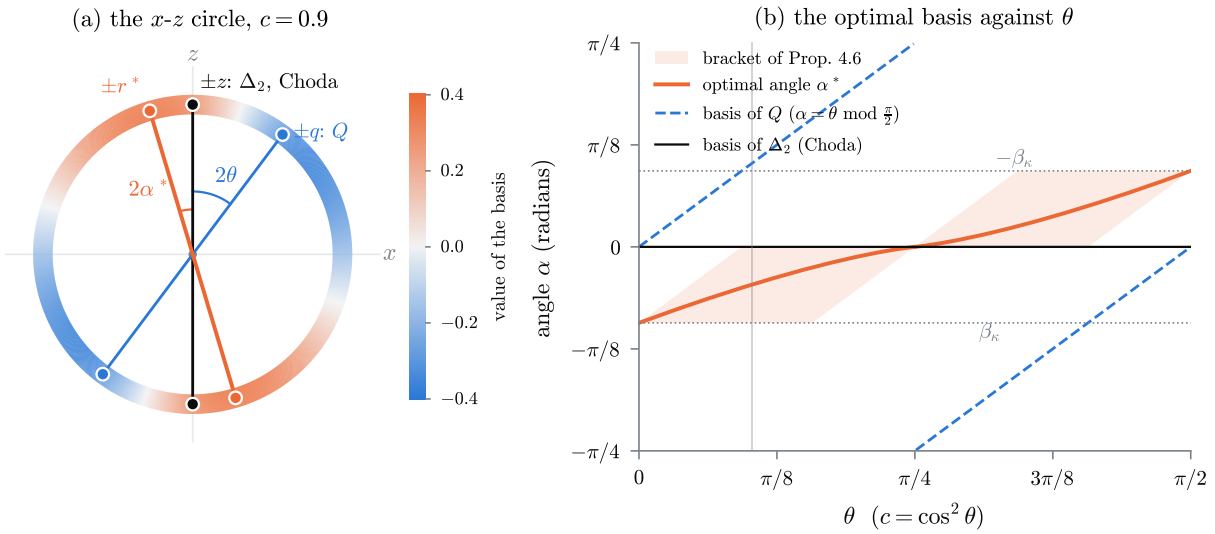


Figure 1: (a) The great circle $r_2 = 0$ of the Bloch ball for $c = 0.9$ ($\theta \approx 0.3218$). An orthonormal basis of $\mathbb{C}^2$ with real coefficients is a pair of antipodal points $\pm r$ at the polar angle $\omega = 2\alpha$ from z, and the ring is coloured by its value $\varphi(\alpha)$ (Theorem 4.4). Black: the basis of Δ_2, i.e. Choda's partition (value $h(U)$); blue: the basis of $Q = U\Delta_2 U^*$ (value $-h(U)$); orange: the optimal basis $\pm r^*$, at the angle $2\alpha^* \approx -0.290$ on the other side of z than q. (b) The optimal angle $\alpha^* \in (-\frac{\pi}{4}, \frac{\pi}{4}]$ against θ, with the bracket of Proposition 4.6 shaded; $\alpha^* \to \beta_\kappa$ as $\theta \to 0$, $\alpha^* = 0$ at $\theta = \frac{\pi}{4}$, and $\alpha^*(\theta) = -\alpha^*(\frac{\pi}{2} - \theta)$. The thin vertical line marks $c = 0.9$.

optimal partitions are exactly the families $x_j = 2t_j\rho_j$ with ρ_j maximising G and $\sum_j t_j\rho_j = \frac{1}{2}$ (Proposition 3.1). Every maximiser r of f has $|r| = 1$ and $r_2 = 0$: as $f(r) = H \geq h(U) > 0$, in Step 2 of the proof we are in the case $|\cos\omega| > |\cos(\omega - 2\theta)|$, where every term of the series is strictly increasing in ϱ, and $r_2 \neq 0$ or $|r| < 1$ would give $\varrho < 1$. Since the maximiser of φ is unique modulo $\frac{\pi}{2}$ (Proposition 4.6), the maximisers of G are the two pure states $\rho_\pm$, and the optimal partition is unique up to splitting $\rho_\pm$ into several multiples (Lemma 2.2(a)).

4.3 The optimal angle

Proposition 4.6. *Let $\frac{1}{2} < c < 1$, i.e. $0 < \theta < \frac{\pi}{4}$, and let β_κ be as in Lemma 4.1(c). Then φ has a unique maximiser α^* in $(-\frac{\pi}{4}, \frac{\pi}{4}]$; it satisfies*

$$\max(\beta_\kappa, \theta - \tfrac{\pi}{4}) < \alpha^* < \min(\beta_\kappa + \theta, 0),$$

and it is the unique solution of $\psi(\alpha) = \psi(\alpha - \theta)$ in this interval. In particular $\alpha^* < 0$, and $\alpha^* \to \beta_\kappa$ as $\theta \to 0$. For $0 < c < \frac{1}{2}$ the maximiser is $\alpha^*(c) = -\alpha^*(1-c)$. For $c = \frac{1}{2}$ the maximiser is $\alpha^* = 0$ (modulo $\frac{\pi}{2}$) and $H = \log 2$.

Proof. Let $I_\ell = (-\frac{\pi}{4} + \ell\frac{\pi}{2}, \ell\frac{\pi}{2})$ and $N_\ell = (\ell\frac{\pi}{2}, \frac{\pi}{4} + \ell\frac{\pi}{2})$, $\ell \in \mathbb{Z}$; by Lemma 4.1(a), $\psi > 0$ on the I_ℓ and $\psi < 0$ on the N_ℓ, all of length $\frac{\pi}{4}$. Let $0 < \theta \leq \frac{\pi}{4}$ and let $W = [\alpha - \theta, \alpha]$ be a window. If W met two of the I_ℓ, it would contain the closure of some N_ℓ together with points on both sides of it, so its length would exceed $\frac{\pi}{4}$; hence W meets at most one I_ℓ, in an interval J of length at most θ. If W meets no positive interval, its integral is non-positive and is smaller than the integral over a window contained in a positive hump. Otherwise choose a window $W' \subset \overline{I_\ell}$ of length θ containing J. Since $\psi \geq 0$ on $\overline{I_\ell}$ and $\psi \leq 0$ on $W \setminus J$,

$$\int_W \psi \leq \int_J \psi \leq \int_{W'} \psi,$$

and the first inequality is strict unless $W \setminus J$ is a null set, i.e. unless $W \subset \overline{I_\ell}$. Hence every maximising window lies in some $\overline{I_\ell}$; modulo $\frac{\pi}{2}$, $W \subset [-\frac{\pi}{4}, 0]$, i.e. $\alpha \in [\theta - \frac{\pi}{4}, 0]$.

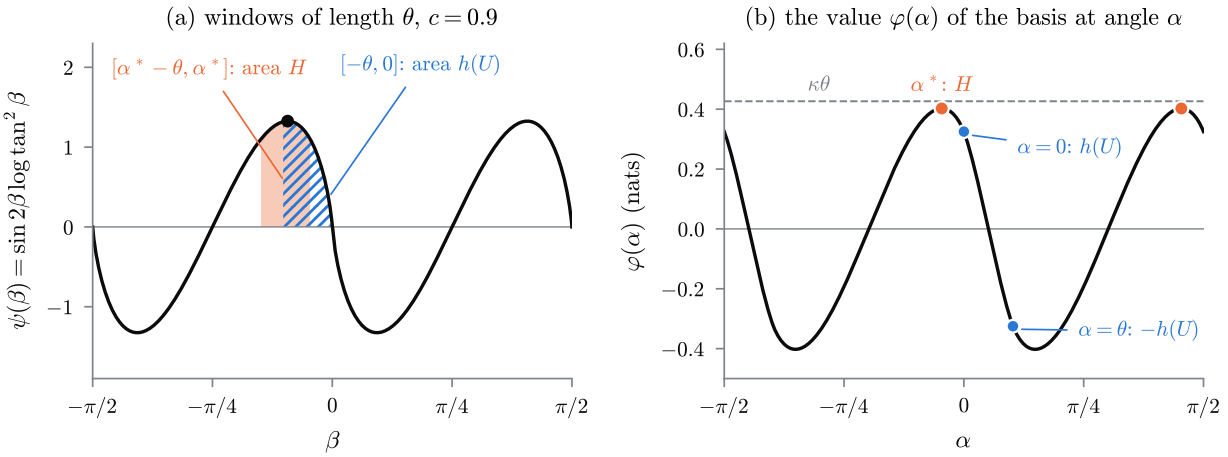


Figure 2: (a) The function ψ for $c = 0.9$. By Theorem 4.4 the value of the basis at angle α is the area under ψ over the window $[\alpha - \theta, \alpha]$. Choda's partition corresponds to the window $[-\theta, 0]$ (hatched, area $h(U)$); the optimal basis to the window of the same length with the largest area, which straddles the maximum $\kappa = \psi(\beta_\kappa)$ (dot) and whose end points have equal values of ψ (shaded, area H). (b) The value $\varphi(\alpha)$ of the basis at angle α, with the three bases of Figure 1 and the upper bound $\kappa\theta$ of Theorem 5.1.

$c = \frac{1}{2}$. For $\theta = \frac{\pi}{4}$ the only such window is $[-\frac{\pi}{4}, 0]$, so $\alpha^* = 0$ and $H = \Psi(-\frac{\pi}{4}) - \Psi(0) = h_2(\frac{1}{2}) - h_2(1) = \log 2$.

$\frac{1}{2} < c < 1$. Now $0 < \theta < \frac{\pi}{4}$. At the end points, $\varphi'(0) = \psi(0) - \psi(-\theta) = -\psi(-\theta) < 0$ and $\varphi'(\theta - \frac{\pi}{4}) = \psi(\theta - \frac{\pi}{4}) - \psi(-\frac{\pi}{4}) = \psi(\theta - \frac{\pi}{4}) > 0$, so neither is a maximiser of the C^1 function φ on $\mathbb{R}$, and every maximiser α^* (modulo $\frac{\pi}{2}$) lies in $(\theta - \frac{\pi}{4}, 0)$ and satisfies $\varphi'(\alpha^*) = 0$, i.e. $\psi(\alpha^*) = \psi(\alpha^* - \theta)$ with $\alpha^* - \theta < \alpha^*$ both in $(-\frac{\pi}{4}, 0)$. As ψ is strictly monotone on $[-\frac{\pi}{4}, \beta_\kappa]$ and on $[\beta_\kappa, 0]$ (Lemma 4.1(c)), two distinct points with equal values lie on different sides of β_κ: $\alpha^* - \theta < \beta_\kappa < \alpha^*$. On the interval $L = (\max(\beta_\kappa, \theta - \frac{\pi}{4}), \min(\beta_\kappa + \theta, 0))$ we have $\alpha \in (\beta_\kappa, 0)$ and $\alpha - \theta \in (-\frac{\pi}{4}, \beta_\kappa)$, so $\psi(\alpha)$ is strictly decreasing and $\psi(\alpha - \theta)$ strictly increasing in α; hence φ' is strictly decreasing on L and has at most one zero there. A maximiser exists by continuity and periodicity, so it is unique modulo $\frac{\pi}{2}$. As $\theta \to 0$, $L \subset (\beta_\kappa, \beta_\kappa + \theta)$ shrinks to β_κ.

$0 < c < \frac{1}{2}$. With $\theta' = \frac{\pi}{2} - \theta$ and Lemma 4.1(b), $\varphi_\theta(\alpha) = \Psi(\alpha - \frac{\pi}{2} + \theta') - \Psi(\alpha) = \Psi(-\alpha - \theta') - \Psi(-\alpha) = \varphi_{\theta'}(-\alpha)$, where φ_θ denotes φ for the angle θ; and $\cos^2 \theta' = 1 - c$. □

The proof is the sliding-window picture of Figure 2: a window of length $\theta \le \frac{\pi}{4}$ that captures the most area under ψ must lie in a single positive hump of ψ, and within the hump it is optimal when the values of ψ at its two ends agree. As $\theta \to 0$ the optimal basis therefore does not approach the basis of Δ_2 (angle 0) or of Q (angle θ), but the basis at the fixed angle β_κ where ψ is largest (Figure 1(b)).

5 Bounds and asymptotics for $n = 2$

Theorem 5.1. *Let $U \in M_2$ be unitary, $c = |u_{11}|^2 = \cos^2\theta$, $\theta \in [0, \frac{\pi}{2}]$, and $\kappa = \max\psi$ as in Lemma 4.1(c). Then*

(a) $h(U) = h_2(c) \le H(\Delta_2 | U\Delta_2 U^*) \le \kappa \min(\theta, \frac{\pi}{2} - \theta)$;

(b) as $\theta \to 0$,

$$H(\Delta_2 | U\Delta_2 U^*) = \kappa\theta - \frac{\kappa + 4/\kappa}{6}\theta^3 + O(\theta^4), \qquad h(U) = 2\theta^2 \log\frac{1}{\theta} + \theta^2 + O\Big(\theta^4 \log\frac{1}{\theta}\Big);$$

in particular $H/h(U) \to \infty$ as $c \to 1$, and likewise as $c \to 0$.

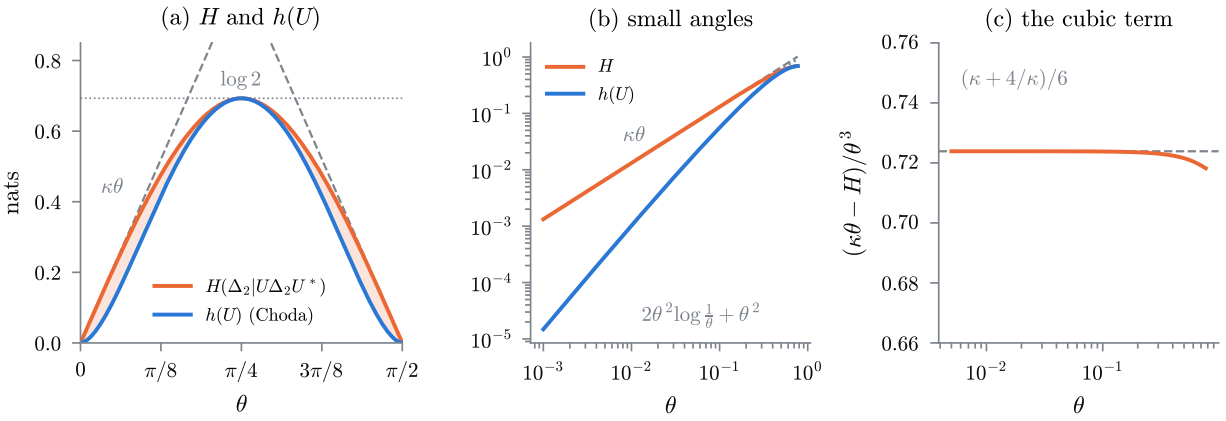


Figure 3: (a) $H(\Delta_2|U\Delta_2U^*)$ (Theorem 4.4) and Choda's value $h(U)$ against θ; the shaded gap $H-h(U)$ vanishes exactly at $\theta\in\{0,\frac{\pi}{4},\frac{\pi}{2}\}$ (Corollary 5.2). Dashed: the bound $\kappa\min(\theta,\frac{\pi}{2}-\theta)$; dotted: $\log 2$. (b) The same on logarithmic axes for small θ, with the leading terms $\kappa\theta$ and $2\theta^2\log\frac{1}{\theta}+\theta^2$ of Theorem 5.1(b). (c) $(\kappa\theta-H)/\theta^3$, which tends to $(\kappa+4/\kappa)/6\approx 0.724$. All curves are computed from the formulas of Section 4.

Proof. (a) The lower bound is $\varphi(0)=h_2(c)$. For every α, $\varphi(\alpha)=\int_{\alpha-\theta}^{\alpha}\psi\le\theta\max\psi=\kappa\theta$, so $H\le\kappa\theta$ by Theorem 4.4(a). Applying this to $1-c$, whose angle is $\frac{\pi}{2}-\theta$, and using Theorem 4.4(c) gives $H\le\kappa(\frac{\pi}{2}-\theta)$.

(b) Fix $\delta>0$ with $[\beta_\kappa-\delta,\beta_\kappa+\delta]\subset(-\frac{\pi}{4},0)$; ψ is smooth there, and by Taylor's theorem there is C with $|\psi(\beta)-\kappa-\frac{1}{2}\psi''(\beta_\kappa)(\beta-\beta_\kappa)^2|\le C|\beta-\beta_\kappa|^3$ for $|\beta-\beta_\kappa|\le\delta$. Let $\theta<\min(\delta,\frac{\pi}{4})$. By Proposition 4.6 the maximising window $W=[\alpha^*-\theta,\alpha^*]$ contains β_κ, so $W\subset[\beta_\kappa-\delta,\beta_\kappa+\delta]$. Since $\int_W(\beta-\beta_\kappa)^2d\beta\ge\theta^3/12$ for every window of length θ (with equality for the window centred at β_κ), $\psi''(\beta_\kappa)<0$ and $\int_W|\beta-\beta_\kappa|^3d\beta\le\theta^4$,

$$\kappa\theta+\frac{\psi''(\beta_\kappa)}{24}\theta^3-C\theta^4\;\le\;\varphi(\beta_\kappa+\tfrac{\theta}{2})\;\le\;H\;=\;\int_W\psi\;\le\;\kappa\theta+\frac{\psi''(\beta_\kappa)}{24}\theta^3+C\theta^4,$$

and $\psi''(\beta_\kappa)/24=-(\kappa+4/\kappa)/6$ by Lemma 4.1(c).

For Choda's value put $s=\sin^2\theta=1-c$, so $h(U)=\eta(s)+\eta(1-s)$. Since $-\log(1-s)=s+O(s^2)$, $\eta(1-s)=s+O(s^2)=\theta^2+O(\theta^4)$. Since $s=\theta^2(1+O(\theta^2))$ and $\log s=2\log\theta+O(\theta^2)$, $\eta(s)=-s\log s=2\theta^2\log\frac{1}{\theta}+O(\theta^4\log\frac{1}{\theta})$. Adding gives the expansion of $h(U)$, and $H/h(U)\sim\kappa/(2\theta\log\frac{1}{\theta})\to\infty$. The case $c\to 0$ follows from Theorem 4.4(c) and the symmetry $h_2(c)=h_2(1-c)$. □

Since $\theta=\arccos\sqrt{c}\sim\sqrt{1-c}$, near $c=1$ the entropy behaves like $\kappa\sqrt{1-c}$, much larger than $h(U)\sim(1-c)\log\frac{1}{1-c}$. The bound $\log 2$ of Proposition 2.3(c) is better than $\kappa\min(\theta,\frac{\pi}{2}-\theta)$ near $c=\frac{1}{2}$. Figure 3 compares H with $h(U)$ over the whole range and checks the two expansions of Theorem 5.1(b), and Table 1 lists values.

Corollary 5.2 (Petz–Szántó–Weiner)**.** *$H(\Delta_2|U\Delta_2U^*)>h(U)$ unless $c\in\{0,\frac{1}{2},1\}$, where $H=h(U)$.*

Proof. This is [17, Appendix]; in terms of Theorem 4.4 it reads as follows. $h(U)=h_2(c)=\varphi(0)$ and $\varphi'(0)=\psi(0)-\psi(-\theta)=\psi(\theta)$, which by Lemma 4.1(a) vanishes exactly for $\theta\in\{0,\frac{\pi}{4},\frac{\pi}{2}\}$. Otherwise 0 is not a maximiser of φ, so $H>\varphi(0)$. For $c\in\{0,1\}$, $H=0=h_2(c)$, and for $c=\frac{1}{2}$, $H=\log 2=h_2(\frac{1}{2})$ (Proposition 4.6). □

Remark 5.3 (Spin model subfactors). Bakshi and Guin use the inequality $H(u\Delta_2u^*|v\Delta_2v^*)\le H(R_u|R_v)$ for the spin model subfactors R_u,R_v of two 2×2 complex Hadamard matrices u,v [1, proof of Theorem 6.22], together with Choda's bound. By Lemma 2.2(b) with $\gamma=\mathrm{Ad}\,u^*$,

c	$h(U)=h_2(c)$	H, Theorem 4.4	$\widehat{H}$, numerical search	α^* (radians)	$\kappa\theta$
0.99	0.056002	0.132043	0.132043	-0.2438	0.132771
0.95	0.198515	0.290618	0.290618	-0.1864	0.298915
0.9	0.325083	0.402396	0.402396	-0.1452	0.426476
0.8	0.500402	0.542607	0.542607	-0.0899	0.614559
0.7	0.610864	0.627949	0.627949	-0.0504	0.768305
0.6	0.673012	0.676658	0.676658	-0.0205	0.907586
0.5	0.693147	0.693147	0.693147	+0.0000	1.041035

Table 1: $H(\Delta_2|U\Delta_2U^*)$ for $c=|u_{11}|^2$: Choda's value, the variational formula of Theorem 4.4 with its optimal angle $\alpha^*\in(-\frac{\pi}{4},\frac{\pi}{4}]$, the maximum over general partitions with four elements found by the numerical search of Appendix A for $U=D_1R_\theta D_2$ with random phases, and the upper bound $\kappa\theta$ of Theorem 5.1.

$H(u\Delta_2u^*|v\Delta_2v^*)=H(\Delta_2|U\Delta_2U^*)$ with $U=u^*v$, so Theorem 4.4 with $c=|(u^*v)_{11}|^2$ raises their lower bound $h_2(c)$ for $H(R_u|R_v)$ to $\max_\alpha\varphi(\alpha)$, e.g. from 0.3251 to 0.4024 at $c=0.9$ (Table 1).

6 Masas of M_n for $n\geq 3$

6.1 A first-order criterion for improving Choda's partition

For $n=2$, Corollary 5.2 was obtained by differentiating the value of a basis at Choda's partition. The same first-order argument works in every dimension. For a unitary $U\in M_n$ let

$$A_U=(u_{jk}\log|u_{jk}|^2)_{j,k=1}^n\quad(\text{with } 0\log 0=0),\qquad M_U=A_UU^*.$$

The map $U\mapsto M_U$ is continuous, since $t\mapsto t\log|t|^2$ is continuous on $\mathbb{C}$.

Proposition 6.1. *Let $U\in M_n$ be unitary, $X\in M_n$ skew-Hermitian, $W_\varepsilon=\exp(\varepsilon X)$, and let $V(\varepsilon)=h(U^*W_\varepsilon)-h(W_\varepsilon)$ be the value (6) of the basis W_ε. Then $V(0)=h(U)$ and*

$$\lim_{\varepsilon\downarrow 0}\frac{V(\varepsilon)-V(0)}{\varepsilon}=-\frac{2}{n}\operatorname{Re}\operatorname{Tr}(M_UX).$$

Consequently, if M_U is not Hermitian, then $H(\Delta_n|U\Delta_nU^*)>h(U)$; an improving direction is $X=\frac{1}{2}(M_U-M_U^*)$, for which the derivative equals $\frac{2}{n}\|X\|_2^2>0$. The set of unitaries with $M_U\neq M_U^*$ is open, and for every $n\geq 2$ it contains $R_\theta\oplus 1_{n-2}$ for $\theta\notin\frac{\pi}{4}\mathbb{Z}$; in particular it has positive Haar measure.

Proof. $V(0)=h(U^*)-h(1)=h(U)$, since $|(U^*)_{ij}|=|u_{ji}|$ and $h(1)=0$.

The term $h(W_\varepsilon)$ is $o(\varepsilon)$. $W_\varepsilon=1+\varepsilon X+O(\varepsilon^2)$, so $|(W_\varepsilon)_{ij}|^2\leq C\varepsilon^2$ for $i\neq j$, and $1-|(W_\varepsilon)_{ii}|^2=\sum_{j\neq i}|(W_\varepsilon)_{ij}|^2\leq C\varepsilon^2$ because the rows are unit vectors. As $\eta(1-\delta)\leq\delta$ for $\delta\in[0,1]$ and η is increasing on $[0,e^{-1}]$, $h(W_\varepsilon)\leq C'(\varepsilon^2+\varepsilon^2\log\frac{1}{\varepsilon})=o(\varepsilon)$ for small ε, and $h\geq 0$.

The term $h(U^*W_\varepsilon)$. Let $b_{ij}(\varepsilon)=(U^*W_\varepsilon)_{ij}=\overline{u_{ji}}+\varepsilon(U^*X)_{ij}+O(\varepsilon^2)$. If $u_{ji}=0$, then $|b_{ij}|^2=O(\varepsilon^2)$ and $\eta(|b_{ij}|^2)=o(\varepsilon)$ as above. If $u_{ji}\neq 0$, $\varepsilon\mapsto\eta(|b_{ij}(\varepsilon)|^2)$ is differentiable at 0 with derivative $\eta'(|u_{ji}|^2)\cdot 2\operatorname{Re}(u_{ji}(U^*X)_{ij})=-2(1+\log|u_{ji}|^2)\operatorname{Re}(u_{ji}(U^*X)_{ij})$. Summing over i,j (the terms with $u_{ji}=0$ may be included, as they vanish),

$$\begin{aligned}\frac{d}{d\varepsilon}\Big|_{0^+}h(U^*W_\varepsilon)&=-\frac{2}{n}\operatorname{Re}\sum_{i,j}u_{ji}(U^*X)_{ij}-\frac{2}{n}\operatorname{Re}\sum_{i,j}(A_U)_{ji}(U^*X)_{ij}\\&=-\frac{2}{n}\operatorname{Re}\operatorname{Tr}(X)-\frac{2}{n}\operatorname{Re}\operatorname{Tr}(A_UU^*X).\end{aligned}$$

$\operatorname{Tr} X$ is purely imaginary, so the first term vanishes, and the second is $-\frac{2}{n}\operatorname{Re}\operatorname{Tr}(M_U X)$.

The improving direction. Write $M_U = M_h + M_a$ with $M_h = \frac{1}{2}(M_U + M_U^*)$ and $M_a = \frac{1}{2}(M_U - M_U^*)$. For skew-Hermitian X, $\operatorname{Tr}(M_h X)$ is purely imaginary and $\operatorname{Tr}(M_a X)$ is real. With $X = M_a$, $\operatorname{Re}\operatorname{Tr}(M_U X) = \operatorname{Tr}(M_a^2) = -\operatorname{Tr}(M_a^* M_a) = -\|M_a\|_2^2$, so the derivative is $\frac{2}{n}\|M_a\|_2^2$, positive if M_U is not Hermitian. Then $V(\varepsilon) > h(U)$ for small $\varepsilon > 0$, and $V(\varepsilon) \le H$.

The example. The set is open because $U \mapsto M_U$ is continuous. For $U = R_\theta \oplus 1_{n-2}$ write $c_\theta = \cos\theta$, $s_\theta = \sin\theta$. The diagonal entries 1 and the zero entries contribute nothing to A_U, and $M_U = M_2 \oplus 0$ with

$$M_2 = \begin{pmatrix} c_\theta \log c_\theta^2 & -s_\theta \log s_\theta^2 \\ s_\theta \log s_\theta^2 & c_\theta \log c_\theta^2 \end{pmatrix} \begin{pmatrix} c_\theta & s_\theta \\ -s_\theta & c_\theta \end{pmatrix} = \begin{pmatrix} c_\theta^2 \log c_\theta^2 + s_\theta^2 \log s_\theta^2 & c_\theta s_\theta \log\cot^2\theta \\ -c_\theta s_\theta \log\cot^2\theta & c_\theta^2 \log c_\theta^2 + s_\theta^2 \log s_\theta^2 \end{pmatrix},$$

which is real, and Hermitian if and only if $c_\theta s_\theta \log\cot^2\theta = 0$, i.e. $\theta \in \frac{\pi}{4}\mathbb{Z}$. A non-empty open subset of $U(n)$ has positive Haar measure. □

For $n = 2$ and $U = R_\theta$ the criterion is exactly Corollary 5.2, so Proposition 6.1 is the n-dimensional form of the argument of Petz, Szántó and Weiner. The next theorem proves non-Hermiticity for Haar-almost every U, independently of the historical searches.

6.2 Almost-everywhere strict improvement

The first-order criterion can be strengthened from a nonempty open set to a full-measure set. We give the analytic and topological details, since the logarithms in M_U are not analytic at zero entries.

Lemma 6.2 (Removing codimension-two sets)**.** *Let N be a connected smooth manifold and let $S_1, \dots, S_r$ be closed smooth embedded submanifolds of real codimension at least two. Then $N \setminus \bigcup_j S_j$ is path connected and dense in N.*

Proof. Each S_j has measure zero in local coordinates, so the complement is dense. We first work in a convex coordinate ball in $\mathbb{R}^d$. Fix two points a, b outside the removed sets. The intermediate points z for which the segment $[a, z]$ meets one of the sets lie in

$$\{a + t(x - a) : x \in S_j,\ t \ge 1\}.$$

After a countable cover of S_j by coordinate patches and of $[1, \infty)$ by compact intervals, this is a countable union of images of C^1 maps of dimension at most $d - 1$ and hence has d-dimensional measure zero. The same applies to segments from b. Thus one can choose z in the ball outside all these exceptional sets; the polygonal path a, z, b misses every S_j. Finally connect any two points of N by a path, cover its compact image by a finite chain of coordinate balls with successive open overlaps, and choose connecting points in those overlaps outside the removed sets by density. The local construction replaces each piece by an avoiding path. Connected smooth manifolds are path connected because they are locally path connected, so this proves the claim. □

Lemma 6.3 (The nonzero-entry unitary domain)**.** *For $n \ge 2$ the open set*

$$\Omega_n = \{U \in U(n) : u_{jk} \ne 0 \text{ for every } j, k\}$$

is connected and has full Haar measure in $U(n)$.

Proof. For fixed j, k, consider the coordinate map $U \mapsto u_{jk}$ with values in $\mathbb{C} \simeq \mathbb{R}^2$. At a unitary with $u_{jk} = 0$, some $u_{j\ell} \ne 0$ with $\ell \ne k$ since row j has norm one. Tangent vectors have the form UK with $K^* = -K$. Set $K_{\ell k} = z$, $K_{k\ell} = -\bar{z}$ and all other entries zero. Then $(UK)_{jk} = u_{j\ell} z$, which ranges over all of $\mathbb{C}$. Consequently the zero-entry set is a smooth closed real codimension-two submanifold. The unitary group is connected, as follows by diagonalisation and a path of eigenvalue phases. Lemma 6.2 gives connectedness of Ω_n. Its complement has Haar measure zero because smooth codimension-two submanifolds have measure zero for the smooth Haar volume. □

Lemma 6.4 (Analytic zero sets)**.** *A nonzero real-analytic function on a connected real-analytic manifold has zero set of measure zero for any measure with smooth density in coordinate charts.*

Proof. At a zero p, not every derivative can vanish: otherwise the analytic Taylor series vanishes on a neighbourhood, and analytic continuation on the connected manifold makes the function identically zero. Choose a nonzero derivative of minimum positive order m at p. Some derivative g of order $m-1$ then satisfies $g(p) = 0$ and $\nabla g(p) \neq 0$. Near p its zero set is a smooth hypersurface by the implicit function theorem. More formally, in a countable atlas the zero set of the original function is contained in the union, over the countably many partial derivatives g, of $\{g = 0, \nabla g \neq 0\}$. Each such regular level set has measure zero. The countable union does too. □

Theorem 6.5 (Generic non-optimality of Choda's partition)**.** *For each $n \geq 2$, for Haar-almost every $U \in U(n)$,*

$$M_U \neq M_U^*, \qquad H(\Delta_n | U\Delta_n U^*) > h(U).$$

Proof. On Ω_n, the function $f(U) = \|M_U - M_U^*\|_2^2$ is real analytic: the unitary entries are real analytic in local coordinates, and $\log|u_{jk}|^2$ is real analytic where $u_{jk} \neq 0$. It is not identically zero. Indeed Proposition 6.1 gives $M_U \neq M_U^*$ at $U = R_\theta \oplus 1_{n-2}$ for $\theta \notin \frac{\pi}{4}\mathbb{Z}$. The map $z \mapsto z\log|z|^2$, extended as zero at $z = 0$, is continuous, so $f > 0$ in a neighbourhood of that example. The dense set Ω_n meets that neighbourhood. Lemmas 6.3 and 6.4 therefore show that $f = 0$ has Haar measure zero on Ω_n, and its complement in $U(n)$ already has measure zero. Apply Proposition 6.1 to every remaining unitary. □

This settles the almost-everywhere lower inequality. It gives neither a strict upper inequality $H < G^*$ nor a separation between general partitions and all orthonormal bases; those require different arguments, examined quantitatively below.

6.3 Numerical findings

For $n \geq 3$ we know no analogue of Lemma 3.2, and we turned to computation. For Haar-random unitaries U (300 for $n = 3$, 150 for $n = 4$, 24 for $n = 5$) we maximised the partition value $\sum_j F(x_j)$ over partitions with n^2 elements, which suffice by Proposition 2.3(b), and separately the value $h(U^*W) - h(W)$ over orthonormal bases W, and G over states, each by multistart local optimisation; whenever a partition appeared to beat the bases, we repeated the basis search with many more starts, including starts built from the directions of the best partition. The method is described in Appendix A.

Status and notation. The analytic criteria above are theorems. The search results that follow are historical numerical observations. Write $\widehat{H}$ for the best value found over general partitions, $\widehat{B}$ for the best value found over bases and $\widehat{G}$ for the best value found over states. The exact values at these admissible points are lower bounds for the corresponding suprema, but the displayed floating-point approximations have no guaranteed error enclosure. Where a comparison depends on it, the points were recomputed with 50 significant digits after normalisation at that precision. None of the searches certifies a global maximum. In particular, $\widehat{H} > \widehat{B}$ does not imply $H > \max_W[h(U^*W) - h(W)]$, and $\widehat{G} > \widehat{H}$ does not imply $G^* > H$. The stored results were not regenerated for this draft.

Table 2 and Figure 4 summarise the results; we describe them in order of increasing departure from the case $n = 2$.

The two searches often agree. For most unitaries the best general partition found is, up to the merging of parallel elements (Lemma 2.2(a)), an orthonormal basis: the best partition and the best basis agree to 8×10^{-12}, 1×10^{-11} and 7×10^{-13} for $n = 3, 4, 5$. For $n = 2$ this is Theorem 4.4.

n	U	starts	partition > basis	largest gap	agreement otherwise	$\widehat{H} - h(U)$: min, median, max			$\widehat{G} > \widehat{H}$
3	300	8/8	22	7×10^{-3}	8×10^{-12}	0.0002	0.0322	0.1333	300
4	150	8/8	6	3×10^{-4}	1×10^{-11}	0.0047	0.0327	0.1762	150
5	24	16/16	2	5×10^{-4}	7×10^{-13}	0.0122	0.0323	0.0849	24

Table 2: Haar-random $U \in U(n)$: the number of unitaries, the starts per unitary for general partitions (n^2 elements) and for bases; the number of unitaries for which a partition beats the best basis found (after the extended basis search) and the largest such gap (both values recomputed with 50 digits); the largest difference between the best partition and the best basis for the others; the excess $\widehat{H} - h(U)$; and the number of unitaries with $\widehat{G} > \widehat{H} + 10^{-6}$. These are comparisons of searched values, not certified global optima.

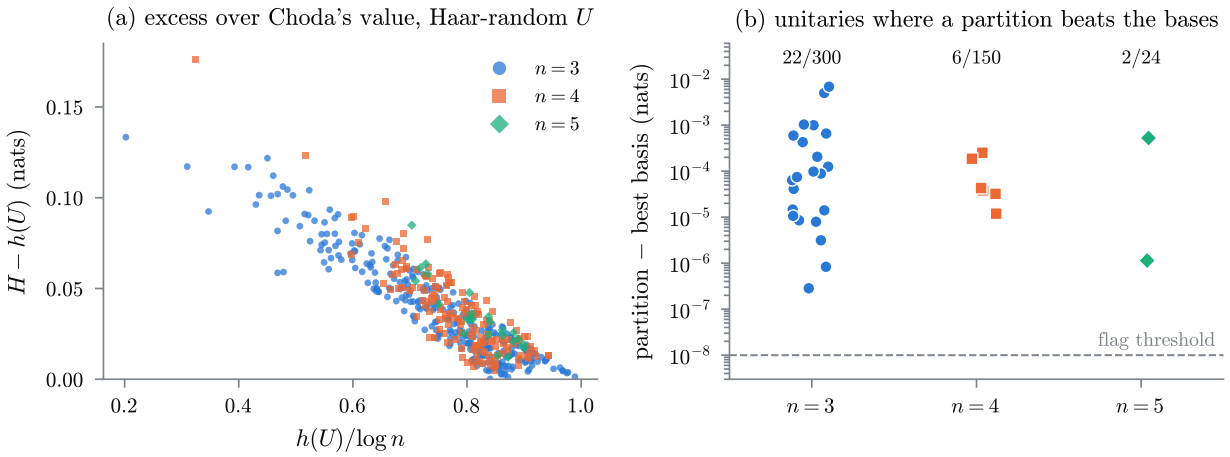


Figure 4: Historical searches for Haar-random unitaries, $n = 3, 4, 5$. (a) The excess $\widehat{H} - h(U)$ against $h(U)/\log n$. The graphic retains the original notation H for the best partition value found, not for a certified entropy. (b) Cases where $\widehat{H} - \widehat{B} > 1 \times 10^{-8}$: both values were recomputed with 50 digits at explicit points. The fractions count such sampled unitaries and do not establish their frequency in Haar measure.

Persistent gaps between searched values. For 22 of the 300 unitaries with $n = 3$, 6 of the 150 with $n = 4$ and 2 of the 24 with $n = 5$, a partition beats the best basis found by more than 1×10^{-8}, and none of these gaps was closed by the extended basis search. The gaps range from 3×10^{-7} to 6.9×10^{-3} for $n = 3$, from 1×10^{-5} to 2.5×10^{-4} for $n = 4$ and from 1×10^{-6} to 5.3×10^{-4} for $n = 5$ (Figure 4(b)). For the unitary U' with the largest gap the values at the two explicit points are

n	partition value	best basis found
3	0.5727784466	0.5659197506
4	1.0949192541	1.0946676581
5	1.1963669266	1.1958404880

The structure of the better partitions. In all these cases the retained elements are numerically classified as rank one using the tolerances of Appendix A, and the elements point in 4, 5 and 6 distinct directions for $n = 3, 4, 5$, that is, in $n + 1$ directions; by Lemma 2.2(a) the partition is, up to the merging tolerance, a partition into $n + 1$ rank-one elements. They are not small perturbations of a basis: two of the directions are far from orthogonal, with the largest overlap $|\langle d, d' \rangle|$ equal to 0.897, 0.980 and 0.991 for the unitaries with the largest gap.

Choda's partition is improved in the sample. In every sample the best value found exceeds $h(U)$, by at least 0.0002 ($n = 3$; Table 2), as predicted by Proposition 6.1 when M_U is not Hermitian. The excess is largest when $h(U)$ is small and vanishes as $h(U) \to \log n$, in accordance with the characterisation of complementary pairs (Figure 4(a)).

The state search lies above the partition search. For every sampled unitary, $\widehat{G} - \widehat{H}$ is at least 0.004, 0.026 and 0.076 for $n = 3, 4, 5$. These searches give lower bounds for two different suprema, so their ordering does not order H and G^*. For the exact three-dimensional example investigated in Section 8, the available analytic bound $H \leq \log 3$ lies above the verified state value and does not decide a strict comparison.

Observation 6.6. In the stored samples for $n = 3, 4, 5$, some explicit partitions with $n + 1$ approximately rank-one directions have larger high-precision values than every basis found by the extended multistart search. This supports, but does not prove, the assertion that non-orthogonal partitions can outperform all orthonormal bases.

Conjecture 6.7. There is a unitary $U \in U(3)$ for which

$$H(\Delta_3 | U\Delta_3 U^*) > \max_{V \in U(3)} [h(U^*V) - h(V)].$$

6.4 The remaining global comparison

Conjecture 6.7 requires an explicit partition value above a global basis upper bound for one unitary. A strict comparison would persist on an open neighbourhood: the fixed partition value and $B(U) = \max_V [h(U^*V) - h(V)]$ are continuous, the latter by compactness of $U(3)$. The new attempts in Section 8 supply an exact rational unitary and a verified lower-bound partition, repeat the basis search independently, and actually test a continuity-based global cover. They do not supply the missing global basis upper bound. The larger number of starts and the high precision of a candidate alone cannot fill that particular logical gap.

Remark 6.8 (Non-abelian subalgebras). The same computation applies to any pair of subalgebras of M_n. For $P = M_2 \oplus \mathbb{C} \subset M_3$ and $P = M_2 \otimes 1 \subset M_4$ against uPu^* for two Haar-random u each, the maxima over partitions with n, $2n$ and n^2 elements agree to 1×10^{-13} and 3×10^{-6}; the historical best partition values are $\widehat{H} = 0.6319, 0.5142$ and $\widehat{H} = 0.6795, 0.6923$, against state-search values $\widehat{G} = 0.6931, 0.6931$ and $\widehat{G} = 0.6931, 0.6931$. For $M_2 \otimes 1 \subset M_4$, $E_P\rho = \mathrm{Tr}_2\, \rho \otimes \frac{1}{2}$ with the partial trace over the second factor, so $S(E_P\rho) = S(\mathrm{Tr}_2\, \rho) + \log 2 \geq \log 2$, and similarly $S(E_Q\rho) \leq 2 \log 2$; hence $G^* \leq \log 2$, a bound approached by the state-search values while the partition-search values remain below it. Neither comparison certifies the exact value of H.

7 Discussion and conclusions

The unrestricted entropy of two masas in M_2 is attained by an orthogonal two-element partition. The reduction is a consequence of the geometry of the full state space, not an assumption about the ranks or commutativity of an optimiser. Antipodal symmetry removes the barycentre constraint in the ensemble representation, and the sign structure of the binary-entropy expansion then moves a maximiser to the relevant great circle. This explains why the family of partitions used by Petz, Szántó and Weiner already contains a global optimum.

The one-dimensional formula also determines the optimiser. For a nontrivial non-complementary pair the optimal angle is unique modulo basis exchange and diagonal phases. The equality-of-endpoints condition for the moving window gives a bracketed scalar equation, avoiding an optimisation over partition lengths. Its small-angle expansion distinguishes the entropy from the restricted value $h(U)$: the former detects the relative position to first order, while the latter loses that term. The mutual-information interpretation in Proposition 2.4 gives the same result an operational reading as the largest difference in information about an ensemble revealed by two projective measurements.

The higher-dimensional first-order criterion has a stronger consequence than the historical searches alone suggested. Its failure set is the zero set of a nontrivial real-analytic function on the connected, full-measure domain of unitaries with no zero entries. Theorem 6.5 therefore

establishes $H(\Delta_n|U\Delta_n U^*) > h(U)$ for Haar-almost every U, in every $n \geq 2$. The proof handles the zero-entry locus explicitly, rather than differentiating logarithms there as analytic functions.

The partition-versus-basis comparison is a different problem. We actually attempted it on the historically largest-gap three-dimensional example. A new rational Cayley construction and scalar Arb evaluation complete the lower-bound half with a four-effect partition of value above 0.572778446636. Independent six-parameter basis searches return about 0.565919750582, but that number remains a lower estimate of the global basis optimum. The performed continuity-cover exploration leaves the upper cap at $\log 3$; the corresponding crude uniform resolution would require approximately 3.44×10^{26} cells. Thus this particular route fails quantitatively at the available budget, despite the stable candidate gap. Section 8 states exactly which point values are certified and which global comparison is not.

The historical figures are retained with their provenance, while the new point verification and covering attempt have separate scripts and inputs. Neither recomputation of a candidate nor compilation of a paper certifies a global maximum or validates every old research run against the corrected library. The qubit formula, its optimality and asymptotics, and the generic strict improvement follow from the analytic proofs; the new pointwise enclosure follows from the independently replayable finite arithmetic check.

8 Attempted certification of the higher-dimensional gap

We now examine the largest recorded three-dimensional gap rather than leaving the proposed comparison as an unworked global-optimisation task. The two objectives are

$$B(U) = \max_{V \in U(3)} [h(U^*V) - h(V)] \leq H(\Delta_3|U\Delta_3 U^*).$$

The required separation is a lower bound for H above a global upper bound for $B(U)$. A high-value partition and a low-value basis do not supply those two halves by themselves.

8.1 An explicit unitary and an independently reconstructed partition

The historical JSON identifies the largest gap at sample 109 and stores its 3×3 unitary, the two searched values, and the rank/direction summary. It does *not* store the corresponding partition vectors or the free matrices from which they were normalised. Consequently the original maximizing partition cannot be recovered from that file alone. We instead reconstructed a new candidate, keeping this new calculation separate from the historical study.

The basis search was implemented independently in NumPy/SciPy, using

$$V = \mathrm{diag}(e^{i\phi_1}, e^{i\phi_2}, 1) R_{23}(\theta_{23}) R_{13}(\theta_{13}, \delta) R_{12}(\theta_{12}),$$

with three angles in $[0, \pi/2]$ and three phases in $[0, 2\pi]$. Column phases are irrelevant to the partition. This is the usual six-parameter description of $U(3)$ modulo column phases, obtainable by complex Givens elimination. A second search uses the first three rows X of a 4×4 unitary, giving the four positive rank-one effects $x_j = X_{:j} X_{:j}^*$ with $\sum_j x_j = I_3$. A six-rotation complex Givens mesh parametrises that unitary modulo its irrelevant column phases. The objective in both searches is evaluated directly from the entrywise probabilities; neither search calls the algebra library or its historical optimiser.

For the final target, 64 basis starts (seed 20261103, at most 1200 iterations each) and 32 frame starts (seed 20261104, at most 1500 iterations each) used L-BFGS-B with analytic derivatives. Central finite-difference checks at an interior point gave maximum gradient errors below 3×10^{-10}. The best values were

$$\widehat{B} = 0.565919750582, \qquad \widehat{H} = 0.572778446637, \qquad \widehat{H} - \widehat{B} = 0.006858696054.$$

The budgets, every final-start value, optimisation status, parameter vectors and matrices are stored in `attempts/basis_gap_results.json`. These are floating-point search results, not certified optima.

8.2 Completing the lower-bound half with exact rational matrices

To obtain an exactly admissible unitary and partition, we converted the numerical matrices to rational Cayley matrices. For a skew-Hermitian matrix K with Gaussian-rational entries,

$$C(K) = (I - K)(I + K)^{-1}$$

is exactly unitary. The supplied file `attempts/rational_point_witness.json` specifies three such matrices: $U_0 = C(K_U) \in U(3)$, a basis $V_0 = C(K_V) \in U(3)$, and $T_0 = C(K_T) \in U(4)$. The skew-Hermitian Cayley parameters were rounded to twelve decimal places with skew symmetry enforced exactly. The final independent search above used the double-precision approximation of this fixed rational U_0. Thus U_0 is a new exact unitary seeded by sample 109, not an assertion that the stored rounded unitary was exact.

Set $X = (T_0)_{1:3,:}$. All entries of U_0, V_0, X are Gaussian rationals and $XX^* = I_3$ exactly. Hence every dephased probability in the partition objective

$$F(X) = \frac{1}{3} \sum_{k,j} [\eta(|(U_0^* X)_{kj}|^2) - \eta(|X_{kj}|^2)]$$

is an ordinary rational. Exact rational matrix identities check admissibility, and Arb encloses only the scalar logarithms in $\eta(t) = -t \log t$. The same procedure encloses the value at V_0 and at the state obtained by normalising the first effect of X.

Proposition 8.1 (Certified point values)**.** *For the exact rational Cayley matrices supplied with this paper,*

$$0.572778446636 < F(X) < 0.572778446637,$$
$$0.565919750582 < h(U_0^* V_0) - h(V_0) < 0.565919750583.$$

In particular,

$$H(\Delta_3 | U_0 \Delta_3 U_0^*) > 0.572778446636, \qquad F(X) - [h(U_0^* V_0) - h(V_0)] > 0.006858696053.$$

For the first normalised frame effect ρ_0, $0.754681285291 < G(\rho_0) < 0.754681285292$.

Proof with a finite computational component. The point checker performs Gaussian-rational matrix inversion and multiplication, checks each skew-Hermitian identity and each unitary identity, and verifies $XX^* = I_3$ and the positivity of the four column weights. Thus the effects are an exactly admissible positive partition and V_0 is an exactly admissible orthogonal basis. It then computes the displayed scalar-entropy expressions with Arb at 256-bit precision and checks strict comparisons against the printed rational decimal endpoints. The positive point gap follows by subtracting the basis upper endpoint from the partition lower endpoint. The checker is `attempts/certify_points.py`; replay requires the standard library and python-flint, while generation additionally uses NumPy. No search result is trusted by this pointwise verification. □

The proposition completes a rigorous lower-bound half for one explicit U_0. It does not prove $F(X) > B(U_0)$: the upper endpoint for the single basis V_0 is not an upper bound for every basis. Likewise, $G(\rho_0) > F(X)$ does not prove $G^* > H$. The universal upper bound $H \leq \log 3 \approx 1.09861229$ is above both $F(X) \approx 0.57277845$ and $G(\rho_0) \approx 0.75468129$, so it separates neither comparison.

8.3 A global upper-bound attempt and its quantitative obstruction

We used the classical entropy-continuity bound to build a global covering attempt on the six basis parameters. Define

$$\omega_3(t) = h_2(t) + t\log 2 \quad (0 \le t \le 2/3), \qquad \omega_3(t) = \log 3 \quad (t \ge 2/3).$$

Lemma 8.2 (A continuity modulus for the basis value)**.** *If $V, V' \in U(3)$ and $\|V - V'\| \le t$, then*

$$|[h(U^*V) - h(V)] - [h(U^*V') - h(V')]| \le 2\omega_3(t).$$

For the six-parameter chart above, one may take t to be the sum of the absolute coordinate changes.

Proof. For unit vectors v, w, their pure-state trace distance is $\sqrt{1 - |\langle v, w\rangle|^2} \le \|v - w\|$. Measurement in any orthonormal basis cannot increase that distance, so the two measurement distributions have total variation at most t. The classical continuity inequality $|H(p) - H(q)| \le h_2(s) + s\log 2$ for distributions on three outcomes at total variation $s \le 2/3$ follows from maximal coupling: couple the outcomes with error probability s, then bound each conditional entropy by $h_2(s) + s\log 2$. For larger distances use $|H(p) - H(q)| \le \log 3$. Apply this to each column, average, and then to the two measurement entropies. Every angle/phase derivative of the displayed unitary factorisation has operator norm at most one; a telescoping product estimate bounds $\|V - V'\|$ by the sum of the six coordinate changes. □

The actual exploratory bisection used 10000 splits, 10001 leaves and 20001 centre evaluations. Its best centre value was 0.528813151. The leaf coordinate-radius sums remained between approximately 1.9635 and 2.1599, where the entropy modulus is saturated. Consequently it did not improve the global cap $\log 3$. The centre evaluations used ordinary floating point; they are not a rigorous interval covering certificate. The rigorous upper bound retained by this attempt is the analytic universal bound $\log 3$.

The scale of the obstruction can be estimated without mistaking it for an impossibility theorem. Even if the best basis found were globally optimal, the continuity allowance would have to be smaller than the observed gap 0.00685869605. Solving $2\omega_3(t) = 0.00685869605$ gives $t \approx 0.00035598$. An equal-coordinate mesh would need half-width about $t/6 \approx 0.000059330$. Across three angle intervals of length $\pi/2$ and three phase intervals of length 2π, this corresponds to about 3.44×10^{26} Cartesian cells. This is an estimate for this crude uniform cover, not a lower bound on the cost of every possible adaptive proof. It explains quantitatively why the performed 10000-split continuity attempt does not close the 0.00686 gap.

Thus the outcome is specific: a genericity question is settled analytically, a new exact partition supplies a verified lower entropy bound, independent basis searches reproduce the candidate gap, and the attempted global continuity cover is too weak at the recorded budget. No global basis upper certificate or theorem that partitions beat all bases is asserted.

A Numerical method

The historical computations for Table 1 (fourth column), Table 2, Figure 4 and Remark 6.8 use our PyTorch [15] library `torch_vn_algebra` [10]; the program is `scripts/connes_stormer.py` of the repository [10], run in double precision (complex128) on a CPU (full run on CPU (4 threads), 726 s in total). The curves of Figures 1–3 and the third and fifth columns of Table 1 are computed directly from the formulas of Section 4 with NumPy [8] and SciPy [23] (`scripts/connes_stormer_figures.py` for the figures).

The functional. Δ_n is the image of the inclusion $\mathbb{C}^n \subset M_n$ with inclusion matrix $(1, \dots, 1)$, and Q_U the image of the same inclusion twisted by U (`inclusions.Inclusion` with `unitary=U`); the conditional expectations onto both are taken for the normalised trace $\tau = \operatorname{Tr}/n$, and $\tau\eta(E(x))$ is computed from the eigenvalues of $E(x)$. The computation is batched and differentiable. The test suite checks it against the plain NumPy formula (5) to 10^{-12} for random partitions with $n = 2, 3, 5$.

Search spaces. (1) General partitions with $m = n^2$ elements,

$$x_j = S^{-1/2} Z_j Z_j^* S^{-1/2}, \qquad S = \sum_{j=1}^{m} Z_j Z_j^*,$$

with free complex $Z_j \in M_n$; every family with invertible S is a partition of unity, and every partition of unity with m elements arises this way ($Z_j = x_j^{1/2}$, $S = 1$). (2) Orthonormal bases, $x_j = w_j w_j^*$ for the columns w_j of a unitary $W = W_0 \exp(X - X^*)$, with value (6). (3) States $\rho = ZZ^*/\operatorname{Tr}(ZZ^*)$ for G^*; the searched value $\widehat{G}$ is the larger of the best state value and the values $G(\rho_j)$ at the normalised elements of the best partition found. It is not an upper bound for H.

Optimiser. `optimize.extremize` runs Adam with the learning rate $0.05 \cdot 0.6^r$ in rounds $r = 0, \dots, 7$, re-centring the unitary chart ($W_0 \leftarrow W$, $X \leftarrow 0$) after every round, and returns per start the best value attained at any iterate; every iterate is an admissible point. All starts for all unitaries form one batch. We used 1200 steps for $n = 3, 4$ and 2000 for $n = 5$, and the numbers of starts per unitary given in Table 2 (16 for the states). The unitaries U are Haar-random, from the library's sampler (QR decomposition with phase correction [12]), with fixed seeds. For $n = 2$ (Table 1) the search over partitions with four elements uses $U = D_1 R_\theta D_2$ with random phases and 16 starts.

Extended basis search. A unitary is flagged if the best partition beats the best basis by more than 1×10^{-8}. For every flagged unitary the basis search is repeated with 64 further starts, half of them Haar-random and half at bases built from the directions of the best partition (the principal eigenvectors of its elements, n of them drawn with probabilities proportional to their traces, perturbed by complex Gaussian noise of size 10^{-3} and orthonormalised by the polar decomposition). For the unitary with the largest remaining gap (for each n) another 1024 starts are run, again half of each kind.

Recomputation in high precision. For every unitary that remains flagged, the best partition and the best basis are recomputed in mpmath with 50 digits: U is replaced by its polar unitarisation $U' = U(U^*U)^{-1/2}$, unitary up to rounding at this precision, and the partition $x_j = S^{-1/2} Z_j Z_j^* S^{-1/2}$ is re-normalised from the stored Z_j (for the basis, $Z_j = w_j$), so that $\|\sum_j x_j - 1\|$ is of the order 10^{-50}. This is a high-precision evaluation, not interval arithmetic. The flagged unitaries are stored in the file `connes_stormer.json` written by the program.

Structure of a partition. Elements of trace below 10^{-7} times the largest trace are discarded; an element counts as rank one if its second eigenvalue is at most 10^{-6} times its trace; the principal eigenvectors d, d' of two elements are merged into one direction if $|\langle d, d' \rangle| > 1 - 10^{-4}$.

Checks. For $n = 2$ the search over general partitions with four elements reproduces Theorem 4.4 to 5×10^{-13}, and the optimal basis of Theorem 4.4(b), evaluated in NumPy, gives the formula to 4×10^{-16} (Table 1). The test suite `tests/test_connes_stormer.py` checks $\kappa = 1.3254868$ to 10^{-6}, compares the formula of Theorem 4.4 with a plain BFGS optimisation (SciPy) over

partitions of M_2 with three elements for four values of c, checks that 300 random partitions with two to five elements never exceed it, that the value does not depend on the phases of U, the integral representation of φ by quadrature, the Bloch reduction of Lemma 4.3, the ensemble identity of Proposition 2.3(a) for $n = 2, 3, 4$, and the agreement of the mpmath recomputation with the double-precision value. For complementary masas (Fourier matrices, $n = 2, 3, 4$) the search gives $H = \log n$ to 1×10^{-10} (Remark 3.3(ii)), and $H(\Delta_3|\Delta_3)$ vanishes to 1×10^{-17}.

Code availability. The library, the programs that compute all numbers, tables and figures, and the tests are available at `https://github.com/andre-a-no/von_neumann_type_i` (MIT licence). The stored data were produced with revision `e8d5873` and have not been regenerated after the numerical corrections listed under `Unreleased` in `CHANGELOG.md`. They remain historical results, not a validation of the corrected implementation; the reproduction status is recorded in `man/REPRODUCIBILITY.md`.

Use of artificial intelligence

OpenAI Codex, a language-model assistant, contributed to restructuring and drafting the manuscript; exploring and developing mathematical arguments, including the almost-everywhere genericity proof; and developing and executing the numerical-search and exact-verification programs. The proofs, computational inputs, checks and limitations are stated in the manuscript and accompanying material. Responsibility for assessing correctness, originality, attribution and the final submission rests with the human authors.

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